

Question

The platinum-containing salt **A** with 40.14% platinum decomposes upon heating to produce elemental platinum, a salt **B**, and a yellowish-green gas **C**. The anion in **A** has an octahedral structure, and its anion reacts with $\text{K}_2\text{C}_2\text{O}_4$ to produce a square planar anion **D**. **D** reacts with a binary compound to form an anion **E**. The mass fraction of Pt in **E** is 59.22%, and it contains a ligand with more than 3 atoms. **E** can dimerize to form a neutral compound **F**.

Which of the following statements is incorrect?

- A.** The cation in **A** corresponds to an element that reacts violently with water.
- B.** The sum of the simplest integer ratios after balancing the chemical equation for the decomposition of **A** is 6.
- C.** Potassium salt of **D** is a red crystal.
- D.** The mass fraction of Pt in **F** is 66% (correct within a 1% error margin).
- E.** All options A-D are correct
- F.** At most three of options A-D are correct.

Answer

Correct Answer: **F**

Detailed Explanation

From the description "yellow-green gas" in the question stem, it can be determined that **C** is Cl_2 .

CHECKPOINT

0.5 PTS

C is Cl_2

From the description "the anion in **A** has an octahedral structure" in the question stem, it can be inferred that the anion is PtCl_6^{2-} . Assuming **A** contains 1 Pt, the molecular weight of **A** can be calculated as 486.05. Subtracting the anion, the cation can be determined to be 2 K^+ . Therefore, salt **A** is K_2PtCl_6 , and salt **B** is KCl .

CHECKPOINT

1 PTS

A is K_2PtCl_6 , **B** is KCl

From the description "the planar quadrilateral anion **D**" in the question stem, it can be inferred that **D** should be PtCl_4^{2-} .

CHECKPOINT

1 PTS

D should be PtCl_4^{2-}

Since Cl in PtCl_4^{2-} is easily substituted, it is highly likely that during the formation of **E**, one Cl^- is replaced by a binary compound or ions from a binary compound. Assuming **E** contains only one Pt , the molecular weight of **E** can be calculated as 329.45. Subtracting one Pt and three Cl , the remaining mass is 28, which should correspond to ethylene. Therefore, **E** is $\text{PtCl}_3(\text{C}_2\text{H}_4)^-$, and the dimerized **F** is $[\text{PtCl}_2(\text{C}_2\text{H}_4)]_2$.

CHECKPOINT

1 PTS

E is $\text{PtCl}_3(\text{C}_2\text{H}_4)^-$

CHECKPOINT

1 PTS

The dimerized **F** is $[\text{PtCl}_2(\text{C}_2\text{H}_4)]_2$

Analyzing the options:

A: Potassium reacts violently with water to release H_2 , which is correct;

CHECKPOINT

0.5 PTS

Potassium reacts violently with water

B: The balanced equation is $\text{K}_2\text{PtCl}_6 = 2\text{KCl} + 2\text{Cl}_2 + \text{Pt}$, with a sum of coefficients of 6, which is correct;

CHECKPOINT

1 PTS

The balanced equation for the decomposition of **A** is $\text{K}_2\text{PtCl}_6 = 2\text{KCl} + 2\text{Cl}_2 + \text{Pt}$, with a sum of coefficients of 6

C: The potassium salt of **D** is K_2PtCl_4 , which is a red crystal;

CHECKPOINT

1 PTS

K_2PtCl_4 is a red crystal

D: The dimerized **F** is $[\text{PtCl}_2(\text{C}_2\text{H}_4)]_2$, with a mass percentage of Pt at 66%, which is correct.

CHECKPOINT

0.5 PTS

The mass percentage of Pt in $[\text{PtCl}_2(\text{C}_2\text{H}_4)]_2$ is 66%

Options A, B, C, and D are all correct, so option E (A-D are all correct) is reasonable.

The question asks for the incorrect statement, so option F is incorrect, and the answer is F.